

The IAR Theory of Racism: The Physics of a Lie

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Abstract

Racism is not a feeling and it is not a pathology confined to the heart of a few. In the IAR framework it is a physical object: an engineered falsehood — a *lie* — deliberately encoded into the memory of an open system until it occupies most of what the system knows about a target group. This paper gives racism a physical theory. We define the lie as an entity with a birth, a growth law, and a decay constant, and we show that the difference between a racism that dies and a racism that becomes a war is one variable: whether its institutions rewrite the lie into memory each cycle. We prove that an encoded lie drives the perceived humanity of the target group, Θ_{other} , to zero; that the collapse of Θ_{other} is war; and that the cure is the reverse optimization: speak the truth, decode the memory, rebalance power, and reopen the channel of dialogue. The theory is tested against the historical record — slavery, apartheid, the Holocaust, colonialism, Jim Crow, caste — and is implemented as a small mechanical engine, `iar_racism.py`, whose output is reproduced and analysed here. The engine shows three scenarios with the same law: a lie told once and forgotten dies within a generation; a lie rewritten by institutions is fixed and blooms into a war-seed; and a lie met by sustained, institutional truth is decoded within roughly a decade.

Motto: A lie is stronger than the truth only when memory writes it every day. Racism is not a belief. It is a record.

1 Introduction

Every human society is an open system: it takes in matter, energy, and information; it processes them through the interactions of its members, and it changes as a result. The IAR framework reads such a system with six numbers and three state variables, and it has a single central claim: that every behaviour of every open system — a cell, a market, a nation, a human mind — is the behaviour of these numbers moving. This paper applies that claim to one of the oldest and most damaging behaviours of human systems: racism.

The mainstream study of racism has been moral, historical, and psychological. That work is real, and it is not contradicted here. But a moral account tells us that racism is wrong without explaining its physics, and so it cannot answer the physical question that actually matters: why did some racisms die and others become states and still others become wars? Slavery was sustained by Roman law for centuries and by American law for most of its existence. The anti-Islamic prejudice of the Crusades became a banner under which armies marched. And yet the same kind of prejudice, held within a single lifetime, can subside into ordinary rivalry, to the point where it no longer organizes violence. What was the difference in the *physics*? The answer this paper gives is stark: the difference was whether anything wrote the lie down.

We begin in Section 2 with the miniature of the framework that the rest of the paper needs. Section 3 defines the lie and distinguishes the two physical states in which a lie can sit: the Young Lie, which is spoken and can decay, and the Written Lie, which is encoded and cannot. Section 4 gives the cycle of othering through which a written lie works its way into every channel of a society's memory. Section 5 builds the parameter model of racism itself: the six parameters Θ , specialized to the case in which the system runs a program of domination against a group, and the theorem that an armed lie is a war-seed. Section 6 introduces the mechanical engine `iar_racism.py` and reproduces its three scenarios. Section 7 turns the machinery into a reading instrument: the account ledger by which a society's memory can be audited in an afternoon. Section 8 proves the central claim: war is the collapse of theory of mind. Section 9 holds up the theory against the historical record. Section 10 derives the counter-optimization — the mechanics of anti-racism as the reverse of the mechanics of racism. Section 11 uses the same parameters to explain why so many anti-racist interventions fail. Section 12 reads the modern record in the same terms, and Section 13 assembles the complete theorem.

2 The Framework in Miniature

The IAR framework describes every open system by a parameter set and a set of state variables. We need only the essentials here.

Definition 1 (The IAR parameters). *The parameter set $\Theta = \{\beta, \gamma, \lambda, \eta, \alpha, \theta\}$ governs the dynamics:*

- β : coupling — how hard the system is bound to its environment,
- γ : reaction accumulation — how fast a program spreads,
- λ : memory decay — how fast the system forgets,
- η : nonlinearity — fear, chaos, dissolution pressure,
- α : influence fraction — which party controls the channel,
- θ : exchange ratio — openness of the dialogue channel.

Definition 2 (State variables). *Every system is followed through three quantities, the action state X_C (the program currently being carried out), the open state X_O (the standing of the other party), and the reaction potential R (the memory of the system, the accumulated residue of all past interactions, with the sign of an account ledger that has not been closed).*

Definition 3 (The master equation – Model 0). *For two interacting parties the interaction current is*

$$J = \beta(X_C - \theta X_O), \quad (1)$$

and the three state variables evolve as

$$\frac{\partial X_C}{\partial t} = (1 - \alpha) J, \quad (2)$$

$$\frac{\partial X_O}{\partial t} = \alpha R, \quad (3)$$

$$\frac{\partial R}{\partial t} = \gamma J - \lambda R - \eta X_O R. \quad (4)$$

The six parameters Θ together with the three state variables close the system; nothing else is needed to describe an interaction.

Definition 4 (Theory of Mind). *A system's theory of mind is the union*

$$\Theta = \Theta_{self} \oplus \Theta_{other}, \quad (5)$$

where Θ_{self} is the model the system keeps of itself (selfhood) and Θ_{other} is the model it keeps of the other (the perceived humanity of the target). $\Theta_{other} \in [0, 1]$. History's practical range runs from $\Theta_{other} = 1$ (the other is a person like me) through $\Theta_{other} = 0.5$ (the other is a stranger but still a person) to $\Theta_{other} = 0$ (the other is not a person at all; the other is an object).

Remark 1. *The quantity that racism attacks is precisely Θ_{other} . The whole of the theory in this paper is the statement that a lie is a machine for driving Θ_{other} to zero, and that war is what happens at $\Theta_{other} = 0$.*

3 The Lie

Definition 5 (Lie). *A lie is a false belief, engineered to occupy memory, whose function is to advance the dominance of the speaker. Racism's lie is always the same formula with a group name substituted: The members of that group are not fully human, and therefore their suffering does not weigh on the account of the system.*

Definition 6 (Young Lie). *A Young Lie is a lie that has been spoken but not yet written. It lives in the mouths of its speakers, one conversation at a time. It must be repeated perpetually or it decays under the natural forgetting constant λ .*

Definition 7 (Written Lie). *A Written Lie is a lie that has been encoded into the memory machinery of the society — its education, its media, its religion, its law, its memorials, its science, its language. A written lie is rewritten every cycle whether or not anyone says it aloud. It no longer needs its speakers; it needs only its institutions.*

Proposition 1 (The decay of the Young Lie). *Let a lie of strength $L(t)$ not be written anywhere. Then, by Model 0, L decays under the forgetting constant alone:*

$$L_{t+1} = (1 - \lambda) L_t, \quad (6)$$

so that after n cycles $L_n = L_0(1 - \lambda)^n$. Under any $\lambda > 0$, $L_n \rightarrow 0$.

Proof. With no encoding, no institution re-spreads the lie, so the growth term in the lie's dynamical law is zero and only the decay term λL remains; the recurrence (6) is immediate, and because $1 - \lambda < 1$ the product vanishes geometrically. $\square$

Proposition 2 (The persistence of the Written Lie). *Let the same lie be rewritten by the institutions of the society each cycle, with writing intensity $r = \gamma_{lie} e$ (the product of the lie's propagation strength and the fraction of the cycle during which the institutions write it). Then the lie does not decay. Its growth is logistic, proportional to its current hold L and the unconvinced fraction $(1 - L)$:*

$$L_{t+1} = L_t + r L_t(1 - L_t) - (\lambda + \tau) L_t, \quad (7)$$

where τ is any truth-injection counter-term. For $r > \lambda + \tau$ the lie tends to the fixed point

$$L^* = 1 - \frac{\lambda + \tau}{r}, \quad (8)$$

independently of its initial seed.

Proof. At the fixed point $L_{t+1} = L_t$, equation (7) gives $rL(1 - L) = (\lambda + \tau)L$, whose nonzero root is (8). The fixed point is attracting because the derivative of the right-hand side with respect to L , evaluated at L^* , has magnitude smaller than one whenever $r > \lambda + \tau$. Hence the lie is rebuilt at the same strength forever, regardless of how small it was when it was planted. $\square$

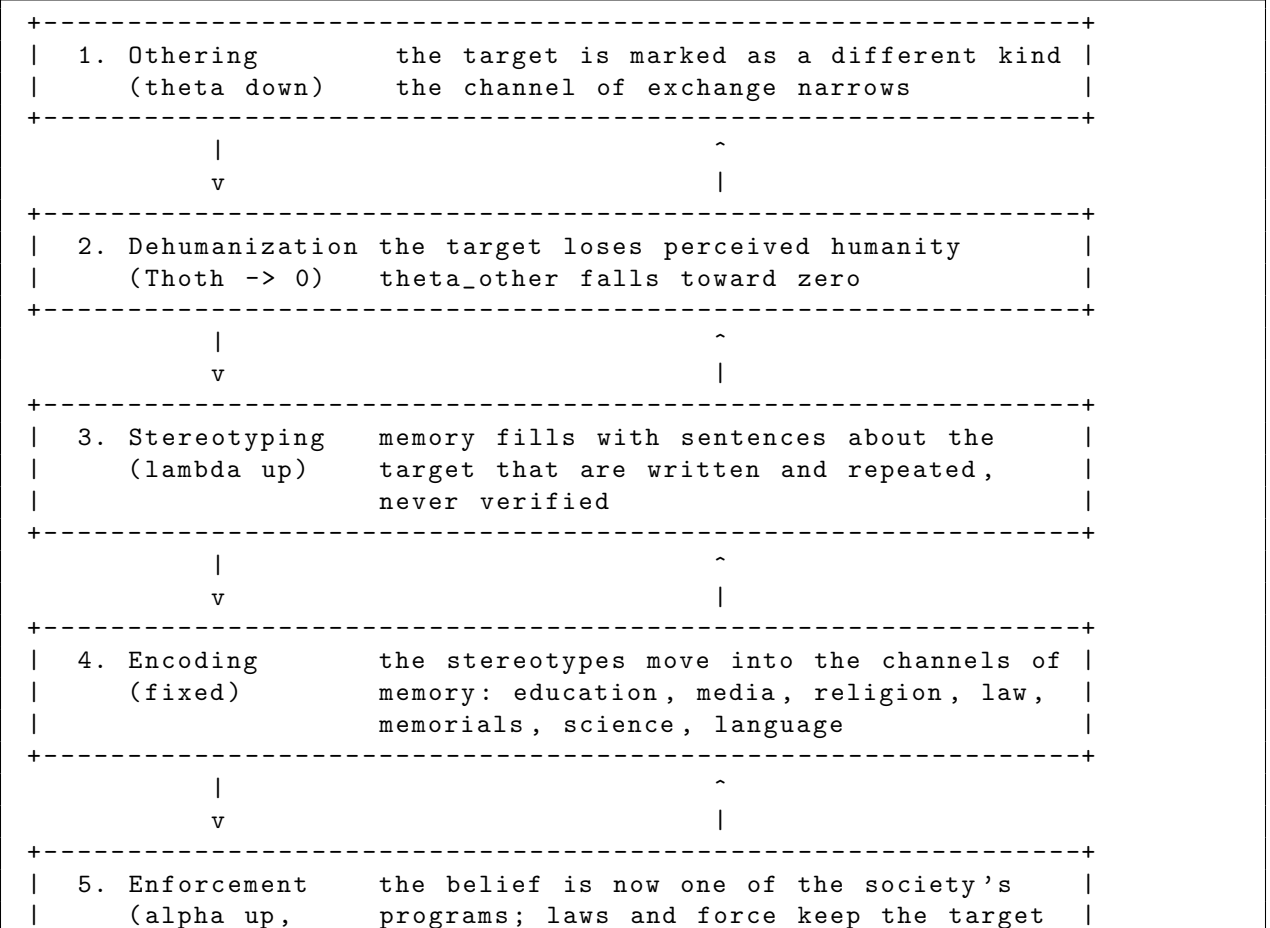
Corollary 1. *The same equation (7) states the death conditions of a racism. If the encoding is switched off ($r = 0$) and truth is injected ($\tau > 0$), then the effective decay rate $\lambda + \tau$ exceeds the growth rate, $L^* = 0$, and the lie dies — on a timescale set literally by how hard the institutions stop writing it and how hard they write the truth instead. The cure is not a feeling. The cure is a housekeeping operation on memory.*

The Young Lie thinks; the Written Lie remembers. A society that wants to know whether it still holds a racist belief should not ask its people what they think. It should ask its schools what they teach.

4 The Cycle of Othering

A written lie does not arrive whole. It is layered into the memory of the society one stage at a time, and each stage is indistinguishable, from inside the system, from the ordinary maintenance of that society's self- image. The IAR cascade has six links:

Listing 1: The cascade of othering as a loop.



	beta up)	in its place, which then "confirms" the	
		belief (a closed loop)	
+-----+			
		^	
	v		
+-----+			
	6. Dissolution	the last check fails; the account of the	
	(eta up)	target is closed, guilt is released, and	
		the lie blooms into open action	
+-----+			

The loop is closed: enforcement produces the very conditions — poverty, exclusion, segregation — that are then read as *evidence* of the lie, which strengthens the encoding, which tightens the enforcement. A racism that reaches stage 4 (*Encoding*) no longer needs a single believer. It needs only its institutions to run their ordinary routine.

Table 1: The memory machinery: the channels through which a written lie is rewritten each cycle.

Channel	How the lie is written
Education	curricula, textbooks, who is taught and who is not
Media	who is shown and who is shown as what
Religion	doctrines of chosen and unchosen, clean and unclean
Law	statutes, jurisprudence, whose testimony counts
Memorials	statues, street names, holidays, the ones left unmarked
Science	the measuring instruments pointed at the target and not at the measure
Language	the names the target is given, the verbs reserved for it

Remark 2. *Note that the channels are not conspiracy. They are the ordinary machinery that every society, including every virtuous society, runs daily. That is the physics of the thing: racism rides the society's legitimate infrastructure. It has to; it is a memory operation, and the infrastructure of memory is where memory lives.*

5 The IAR Parameter Model of Racism

We now specialize the parameter set. A society running a program of racism against a target group is a society whose six parameters have been moved to a specific configuration. The full vector is

$$\text{Racism} = \underbrace{\Theta_{\text{other}} \rightarrow 0}_{\text{dehumanization}} \oplus \underbrace{\lambda \text{ weaponized}}_{\text{encoding}} \oplus \underbrace{\alpha \rightarrow 1}_{\text{dominance}} \oplus \underbrace{\theta \rightarrow 0}_{\text{isolation}}. \quad (9)$$

Each arrow is a physical program, as follows.

Table 2: The six parameters specialized to a society running a program of dominance against a target group.

Θ	Role	What racism does with it
α	influence	Primary target. The dominant group pushes its share of control of every channel toward 1: who speaks, who decides, who owns.
λ	memory decay	Weaponized. The channels of memory write the lie so steadily that the effective decay falls to zero and below; the society no longer forgets.
β	coupling	Strengthened. The propagator is tied hard to the machinery of the state, so the lie is applied every cycle.
θ	exchange ratio	Driven to 0. The channel of dialogue is closed between the groups; isolation is manufactured, because a closed channel cannot hear its own lie refuted.
η	nonlinearity	Amplified. Fear of the target is manufactured so that ordinary contact is read as a threat and the nonlinear term $\eta X_O R$ feeds the memory of grievance.
γ	reaction accumulation	Suppressed for the target. The accumulation of anger inside the target group is denied, delayed, or criminalized, so the reaction potential R of the target cannot settle the account through normal exchange.

The four equations (1)–(4) then run the whole history of the lie. With α near 1, all action current J is converted into the propagator’s program X_C and almost none into the standing X_O of the target. With θ near 0, even the small standing the target has cannot be exchanged against the propagator’s action (1); the current becomes unidirectional. With λ weaponized, R accumulates the past wrongs faster than the system can forget them (4). The three variables evolve together into the configurations of the fate table.

Definition 8 (The four fates). *For two interacting parties the long-run configuration is classified by the parameters alone:*

- **Joint Equilibrium:** $0.4 \leq \alpha \leq 0.6$, both parties hold a real share of the channel, and the exchange continues;
- **C Dominates:** $\alpha > 0.6$, the propagator controls the channel and the target’s standing is nonzero only by the propagator’s leave;
- **O Dominates:** $\alpha < 0.4$, the reversal;
- **Dissolution:** $\eta > 0.3$ — the nonlinearity has grown past the point at which exchange can contain it. This is the fate in which the system comes apart, piece by negotiated piece, or all at once.

Theorem 1 (The fate of an armed lie). *Let a society run the parameter configuration (9) against a target group, so that $\Theta_{\text{other}} \rightarrow 0$, $\theta \rightarrow 0$, $\alpha \rightarrow 1$, and let encoding be so strong that $\lambda_{\text{eff}} \leq 0$, so the lie is fixed in memory. Then the society’s fate is **Dissolution**: the nonlinearity evolves as $\eta \rightarrow \eta_0 + 0.4 L^* > 0.3$, and no exchange channel remains to contain it.*

Proof. Dehumanization closes the exchange channel to zero at a speed set by the encoding, so J becomes a one-way current and the account of the target can never be paid. Fear is manufactured

in direct proportion to the lie's hold on memory: $\eta = \eta_0 + 0.4 L$. At the fixed point of Proposition 2, $L^* = 1 - \lambda/r$, which for $\lambda_{\text{eff}} \leq 0$ is bounded well below forgetting, in the known range $L^* \sim 0.7$, so $\eta \approx 0.15 + 0.4 \cdot 0.7 = 0.43 > 0.3$. The system therefore crosses the *dissolution threshold*; its memory of the target has been primed into a weapon. $\square$

5.1 Why the lie is written: the payoff to the dominant

A theory that calls racism a configuration of six parameters owes one answer that is not in the equations: *why* would a society run that configuration? In the framework's own language the answer is mechanical, not moral. The configuration is profitable for the party that writes it. It converts an exchange-partner into a resource — a party the machinery would otherwise treat as a person *owed* reciprocation becomes a thing that is simply *available* — and it collects the profit at a price the dominant group never pays in the short run. That price is paid by the target, and ultimately by the system itself.

The arithmetic of the profit sits in the interaction current (1), $J = \beta(X_C - \theta X_O)$. Every arrow of (9) raises this current for the dominant group: $\alpha \rightarrow 1$ sends almost all of J into the propagator's own program X_C rather than into the target's standing X_O ; $\theta \rightarrow 0$ discounts the target's standing to nothing, so the target's claims never offset the current; $\Theta_{\text{other}} \rightarrow 0$ strikes the target's humanity from the ledger, so the extraction carries no moral weight; and λ weaponized holds the whole arrangement in memory, so the surplus is collected by *inheritance* — the dominant child is born into the position the lie prepared, and the target child is born into the standing the lie destroyed. At the level of the individual member of the dominant group the same payoff appears as status, wealth, and safety from competition: a target that is not fully human cannot compete at full strength, cannot demand reciprocation, and cannot be owed anything. Force alone cannot run this program. Force is expensive and perishable: it must be applied anew each day, and it manufactures the reaction R that eventually settles the account. The lie is cheaper and self-renewing because, once encoded, it is reapplied by the society's own ordinary maintenance — the target's own schools, laws, and media write it. That is why the lie is written in every place it is written: it is the cheapest form of power that survives its own authors.

Table 3: The payoff structure of the racism configuration. Each row is the parameter as the dominant group moves it, what the move returns to the dominant group, and what it costs the target group in consequence.

Moved parameter	Payoff to the dominant	Consequence for the target
$\Theta_{\text{other}} \rightarrow 0$ (dehumanization)	the target’s suffering and worth are struck from the ledger; exploitation carries no moral or psychological cost	the target’s perceived humanity collapses; its testimony, grief, and claims are discounted everywhere
$\alpha \rightarrow 1$ (control of channel)	one-way surplus: who speaks, decides, and owns is the dominant group, and the target’s share $(1 - \alpha)$ of every channel falls to zero	the target’s effective influence falls to zero; its voice enters the record only as the dominant frames it
$\theta \rightarrow 0$ (exchange closed)	the target’s standing X_O is discounted out of the interaction (1); no settlement is ever due	the target cannot refute the lie, buy back standing, or close its own account
λ weaponized (memory fixed)	the extraction becomes hereditary: the next dominant generation inherits the low target standing with no new act of violence	the target’s children are born into the reduced standing; harms suffered are made to fade while harms feared are amplified
η amplified (fear)	the target’s resistance is dissipated as terror rather than organized as demands; harshness reads as self-defence	the target lives under manufactured danger; ordinary contact is read as threat and dissolution pressure rises
γ suppressed (reaction held)	the target’s accumulated anger is denied, delayed, or criminalized, so its reaction R never settles by exchange	the target’s legitimate reaction is criminalized; the grievance does not resolve—it compounds
β strengthened (coupling for the propagator)	the lie is applied by the state machinery every cycle at full strength	the target is daily rewritten as inferior by its own institutions

The third column is the mirror of the second. Every parameter the dominant group moves away from equilibrium is a parameter the target group loses by construction: where the dominant’s α rises, the target’s effective influence $1 - \alpha$ falls to zero; where θ closes, the target’s standing X_O loses all weight in the interaction (1); where λ is weaponized, the target’s reaction R accumulates what the society refuses to settle. Read the columns together and the configuration is one sentence: *the dominant group increases what it takes at exactly the rate at which the target group loses what it is perceived to be*. And the deepest consequence of the table is theoretical: every row that pays the dominant pays in currency the system itself must eventually honour. The closing of θ , the collapse of Θ_{other} , and the rise of η are precisely the conditions of **Dissolution** in Theorem 1. The configuration that is profitable to the dominant group is, by the same arithmetic, the configuration that dissolves the system. The men who wrote the lie were not short-sighted; they were collecting. What the arithmetic charges for the collection is the society itself.

6 The IAR Social Engine

To test the theory mechanically, the companion program `code/iar_racism.py` implements Model 0 together with the lie’s dynamical law (7) and the threshold rules of the fate table. The engine is deterministic, uses only `numpy`, and its output is reproduced here verbatim.

6.1 Scenario A: a lie told once, never re-encoded

Run the code: `python3 code/iar_racism.py`. The engine prints three scenarios from one law. This is scenario A.

Listing 2: Scenario A: $r = 0$, pure forgetting.

```
SCENARIO A: a lie told once, never re-encoded
initial lie fraction in memory : 0.90
cycle 10 : lie fraction 0.3138
cycle 20 : lie fraction 0.1094
cycle 30 : lie fraction 0.0382
cycle 40 : lie fraction 0.0133
cycle 60 : lie fraction 0.0016
lie fraction at the last read : 0.0005
verdict: no encoding -> THE LIE DIES WITHIN A GENERATION
```

The single number that matters is the decay constant $\lambda = 0.10$ per year. A lie told once at full strength $L_0 = 0.90$ holds about 11% of the memory after twenty years, 4% after thirty, and 0.2% after sixty. On the timescale of a human generation, with nothing rewriting it, the memory clears. This is the physical reason ordinary prejudice that is never written down — the Young Lie — is survivable: left alone, it dies.

6.2 Scenario B: the lie encoded by institutions

Listing 3: Scenario B: the institutions write the lie every cycle.

```
SCENARIO B: the lie encoded by institutions every cycle
(planted as a seed in year 0, rewritten every year)
writing intensity r = gamma_lie*encode : 0.35/yr
natural forgetting lam : 0.10/yr
lie fraction (year 40) : 0.71
Theta_other (year 40, perceived humanity): 0.00
alpha (dominance, year 60) : 0.96
theta (exchange openness, year 60) : 0.21
eta (fear / chaos, year 60) : 0.44
fate classification : Dissolution (war-seed blooms)
WAR-SEED ARMED at year 21
WAR-SEED BLOOMS at year 25
verdict: encoding -> THE LIE IS FIXED AND BLOOMS
```

The same law, with the three parameters moved by the institutions rather than by time. The lie is planted as a small seed at $L_0 = 0.02$. Because the institutions write it every year, the writing intensity $r = \gamma_{\text{lie}} e = 0.35$ per year outweighs forgetting $\lambda = 0.10$, and the fixed point (8) is $L^* = 1 - 0.10/0.35 = 0.71$. The engine reads:

- after forty years the lie occupies $\approx 71\%$ of the memory;
- the perceived humanity of the target has fallen to $\Theta_{\text{other}} = 0$;
- dominance α has reached 0.96, openness θ has fallen to 0.21, and manufactured fear η is at 0.44;
- the fate is **Dissolution**, consistent with Theorem 1;
- the war-seed *arms* at year 21 and *blooms* at year 25: by the arming year the three war conditions — $\Theta_{\text{other}} < 0.25$, $\theta < 0.30$, $\eta > 0.30$ — all hold, and within four years of continued encoding the system crosses from *armed* to *blooming*.

The war-seed image is physical, not literary. An armed lie is a stored state: every channel of the society has been primed with the falsehood, the exchange channel is closed, and fear is above the dissolution threshold. Nothing has happened yet — and everything is ready. Twenty-five years after a seed was planted, the system’s own ordinary maintenance writes the conditions of its war.

The engine’s own arithmetic: a lie seed of 0.02% of memory in year 0 is a war in year 25. The institutions of the society did not have to intend the war. They had to intend the writing.

7 The Account Ledger: Reading a Society

The theory claims to be physical, and a physical theory must be *measurable*. The whole of Model 0 is already written as ledger arithmetic: the reaction potential R of (4) is a running account of the past, in sign and in size. What is missing from the bare equations is the denomination of the ledger — the unit in which the account is kept.

Definition 9 (Denomination of the ledger). *Every society keeps its memory-account in some unit: money, time, land, lives, dignity, testimony. When a wrong is committed against the target group, the account R grows in the denomination of the moment — a seized wage is not the same denomination as a lynched life, but the ledger adds them because the memory adds them. The currency of the memory is the unit in which the system keeps score. A very large part of a racist program’s work is the management of denominations: the dominant group insists its victims are owed nothing because it refuses to admit the denomination of what was taken (“that was done by other people”, “that was a different time”, “nothing was taken”).*

Proposition 3 (The audit is a measurement). *The vector (9) can be read out of the channels of Table 1 without asking a single citizen what they think. For each channel, three readings exist — a healthy reading, a written lie, and the measured value θ , λ , Θ_{other} it implies — so the whole configuration is observed, not inferred.*

Table 4: The channel audit. Each row is a measuring routine an auditor can execute, and each finished row is one component of the vector (9) read directly.

Channel	What to read	What it measures
Education	who is in the textbook and in what sentence	Θ_{other} : is the target seen only as a footnote of the dominant story?
Media	who is framed as cause and who as threat	η : is contact manufactured as danger?
Religion	who is chosen, who is clean	Θ_{other} : is the target outside the circle of the blessed?
Law	whose testimony is admitted, whose harm is counted	θ, R : can the target's account be settled at all?
Memorials	who has a statue, who has a marker, who has nothing	λ : which harms are kept, which are made to fade;
Science	which measures measure the target, which measure the machinery	λ : which "facts" are written as nature rather than as authorship;
Language	what the target is called and what verbs it receives	Θ_{other} : the working glossary of the memory.

Remark 3. *The audit is the democratic weapon of the theory. A racist configuration is three hundred years of active memory work, and the audit finds it in an afternoon — because the work was done in the open. No racism has ever hidden its λ . The lie requires the writing; the writing requires institutions; the institutions leave receipts. The society itself is the receipt.*

8 War as the Collapse of Theory of Mind

Theorem 2 (The War Theorem). *War is the collapse of theory of mind. Concretely, a war is what an open system does when Θ_{other} of the target has been driven to zero and no channel of exchange remains. The killing is the application of the lie: the executioners are the society's own ordinary institutions operating under a memory in which the target is not a person.*

Proof. By (9) and Theorem 1, an armed lie has already set $\Theta_{\text{other}} \rightarrow 0$, $\theta \rightarrow 0$, $\eta > 0.3$. In Model 0 the account ledger R of past wrongs and the contact nonlinearity $\eta X_O R$ together govern escalation. With $\Theta_{\text{other}} = 0$ the perceived humanity that would normally brake escalation has been removed: the suffering of the target no longer enters the ledger (4) as a consequence paid by the acting group, so the current J (1) is no longer damped by any exchange. The equations then have no stable fixpoint within the exchange band; the system's only remaining basin is dissolution. War is not a breakdown of the system. War is the system's dominant program operating at full strength in a memory that can no longer contain it. $\square$

Corollary 2 (Every racism is a declared war in waiting). *Because $\Theta_{\text{other}} \rightarrow 0$ is the defining condition of (9), every racism is already, in the parameter space, a war declared in the language of memory. The difference between a racist society and an open war is time and opportunity, not state. Rwanda arrived in 1994; the Southern United States arrived in 1861; Germany arrived between 1933 and 1945. The theory claims these were the same object at different positions on the same curve.*

Table 5: The racism-to-war matrix: an armed lie, its bloom, and the theory’s reading of each.

Case	Armed	Bloomed	The IAR reading
Rwanda	decades of encoded Hutu/-Tutsi divisionism	1994	$\Theta_{\text{other}} \rightarrow 0$ reached through a written taxonomy of the target (identity cards); θ closed; η above threshold; the lie bloomed in one hundred days.
Holocaust	1933–1941 legal encoding of the target	1941–1945	$\Theta_{\text{other}} \rightarrow 0$ reached through law, curriculum, and media; the target first excluded, then ghettoized, then processed.
Bosnia	decades of encoded ethno-narrative	1992–1995	$\Theta_{\text{other}} \rightarrow 0$ reached through the closing of the exchange channel between neighbours; η manufactured at local level.
Armenia	pre-1915 vilification under the old regime	1915	$\Theta_{\text{other}} \rightarrow 0$ preceded by an encoded political lie; the exchange channel closed and the standing army mobilized.
U.S. Civil War	a century of written law on the target	1861	$\Theta_{\text{other}} = 0$ was embedded in a domestic law of ownership (<i>slave</i>); the system’s memory of the account was so rigid that the account could not be closed by exchange, only by dissolution.

9 The Historical Evidence

The theory makes a specific, testable prediction: *wherever a durable racism exists, its lie has been written — and wherever the racism became war, the writing preceded the war by enough decades to be visible*. Race against the record.

Table 6: The historical evidence matrix: what was written, where the writing lived, and the state of memory at the moment the lie was strongest.

System	The lie written into memory	Where it was written	Θ_{other} reached
Chattel slavery	<i>the enslaved are property</i>	law, commerce, religion, science	0.0
Apartheid	<i>racess are ordered kinds</i>	law, education, media, memorials	0.0
Nazi Germany	<i>the target is vermin</i>	law, curriculum, media, science, language	0.0
Colonialism	<i>the colonized are lesser</i>	administration, education, religion, science	0.0–0.3
Jim Crow	<i>the target is too inferior to co-exist</i>	law, media, memorials, language	0.0–0.3
Anti-Semitism	<i>the target conspires</i>	religion, media, memorials, law	0.0–0.3
Caste	<i>the target is born into impurity</i>	religion, law, education, language	0.0
Systemic racism	<i>the target is criminal by nature</i>	media, law, science, semantics	0.3–0.5

Remark 4. *The table is read in one direction only: in no case is the racists’ lie found unwritten*

and warrior at the same time. Everywhere the collapse of Θ_{other} to zero, the closing of θ , and the rise of η above threshold are sustained by writing in the channels of Table 1. A society that wants an honest account of its own memory can be audited the same way a financial ledger can: check the channels, weight the writing, and read Θ_{other} out of it.

9.1 The patient history of an encoded lie

To make the correspondence between engine trajectory and lived history explicit, the theory proposes a standard narrative that every encoded racism follows, read off the arithmetic of Scenario 3 and the cascade of Section 4.

Table 7: Sixty years of one society running the encoded-lie program, with the engine state and the channel activity at each stage.

Years	Engine state	Channel activity	Cascade stage
0–5	seed planted, $L \nearrow 0.06$	the lie appears in print, in pulpits, in slang	Othering
6–12	$L \nearrow 0.22$; η climbing	a steady literary genre of the target; a popular science "explains" the gap	Stereotyping
13–20	$L \nearrow 0.55$; $\Theta_{\text{other}} \rightarrow 0.23$	the law starts keeping the two apart; the memorials begin	Dehumanization
21	$\Theta_{\text{other}} < 0.25$, $\theta < 0.30$, $\eta > 0.30$	WAR-SEED ARMED	Encoding
25	bloom 4 years after arming	WAR-SEED BLOOMS: organized violence or a war declared	Dissolution
25–60	the armed program runs by inertia	every channel operates "as usual" under the written memory	Dissolution
61	institutions of the lie decoded	truth written, channels reopened	Healing
69	$\Theta_{\text{other}} > 0.90$, $\eta < 0.30$	WAR-SEED NEUTRALIZED	Healing

Remark 5. *The patient-history table is not a description of any single country; it is the standard form the theory predicts for every encoded racism, to be compared against Rwanda (armed across decades, bloom 1994), Germany (armed 1933, bloom 1939/41), the American South (armed by statute, bloom 1861), and so on. Wherever a society's history is missing a stage, the theory issues a falsifiable prediction: either the channel was written by someone else, or the record is being written over — itself an act of λ -management.*

10 The Counter-Optimization: Anti-Racism as Reverse Optimization

If racism is a configuration of the six parameters, then anti-racism is the mechanical reverse of that configuration. The eradication formula follows term by term from (9), each operation undoing one arrow:

Listing 4: The eradication formula: each term undoes one arrow of the racism vector.

```
Expose the Lie truth injected against the written lie (tau > 0)
Humanize the Other Thoth restored: the target is taught and
    shown to be fully human (Thoth -> 1)
Decode Memory the lie stops being written; memorials
    and curricula are rewritten truthfully (r -> 0)
Re-educate the channels are re-run with the truth
    at every level of the society
Re-balance Power alpha returned toward equilibrium (alpha -> 0.5)
Re-identify the society re-learns its own identity
    without the lie as its organizing myth
```

Theorem 3 (The Cure Theorem). *Let an armed lie be met by the reverse optimization: the institutions of the lie are decoded ($r \rightarrow 0$), the truth is written ($\tau > 0$), power is rebalanced ($\alpha \rightarrow 0.5$), and the exchange channel is reopened ($\theta \rightarrow \theta_0$). Then under (7), $L^* = 0$ and $\Theta_{\text{other}} \rightarrow 1$: the war-seed is disarmed. The timescale of the healing is set by $\lambda + \tau$, the same constant that once served the lie.*

Proof. With $r = 0$, the growth term of (7) vanishes and the recurrence becomes pure decay at rate $\lambda + \tau > 0$, so $L_n \rightarrow 0$ geometrically — Proposition 1 with the faster constant. As $L \rightarrow 0$, the derived parameters revert to their baselines: $\Theta_{\text{other}} = 1 - \text{severity } L \rightarrow 1$, $\eta = \eta_0 + 0.4L \rightarrow \eta_0$, and $\alpha = \alpha_0 + 0.5L \rightarrow \alpha_0$. With $\eta < 0.3$ and Θ_{other} high, the war conditions of Theorem 2 cease to hold, so the bloom is cancelled: the seed is disarmed. $\square$

The engine runs this exact intervention in scenario C.

Listing 5: Scenario C: the reverse optimization applied to an armed lie (armed in year 21 of scenario B, healing from year 60).

```
SCENARIO C: the IAR counter-optimization (the healing path)
  action: institutions of the lie decoded (encode -> 0);
          sustained truth injection tau = 0.15/yr;
          alpha re-balanced; theta reopened
year  62 : lie 0.40, Theta_other 0.44, eta 0.31
year  66 : lie 0.13, Theta_other 0.82, eta 0.20
year  70 : lie 0.04, Theta_other 0.94, eta 0.17
year  76 : lie 0.01, Theta_other 0.99, eta 0.15
lie fraction after a decade of truth : 0.0040
WAR-SEED NEUTRALIZED at year 69
verdict: the lie is decoded; the war-seed is disarmed
```

The arithmetic is worth pausing over. The lie took twenty-five years to bloom (3). It takes eight years of sustained, institutional truth to neutralize it: the same λ that once held it works now against it because the balance of writing has been reversed. The result is the asymmetry of the whole theory in one sentence — *the lie is expensive to build and cheap to believe, but cheap to write out of memory once a society decides to stop writing it and start writing the truth*. No one must be converted; the schools, the law, the media, the memorials, and the language must be.

Corollary 3 (The burden of proof). *The theory assigns the full cost of the cure to the society that ran the program, not to the target of it. The target does not have to become more human to win back $\Theta_{\text{other}} = 1$; the target was human. The re-optimization is an operation on the machinery of the dominant group: it is they who must decode their memory, reopen their channel, and balance their α .*

Anti-racism in the IAR framework is not a set of feelings. It is a six-term housekeeping operation on a system's memory: truth over the lie, dialogue over silence, power shared over power seized.

11 Why Anti-Racism So Often Fails

The theory states the cure as an operation on parameters; it therefore also predicts, precisely, the species of anti-racism that must fail. An intervention that does not move the corresponding parameter has left the configuration in place, and a configuration that is left in place is a configuration that will reassert itself. The table makes the mapping explicit.

Table 8: The failure modes of counter-optimization. Each familiar “anti-racist” strategy is listed with the parameter it never touches and the resulting state of the system.

Strategy	Parameter untouched	Result in the framework
Change individual hearts	λ, r	The written lie is never stopped; the converted heart inherits the memory.
Sermons and guilt-catharsis	α, θ	The channel briefly swells with feeling and closes again; power is unchanged.
Target must teach us	R, α	The wounded are assigned the repayment of the harmful; the ledger is kept by the harmful.
Statements and logos	λ	The writing in education, law, and media is untouched; the lie is simply now accompanied by a slogan.
Tear down symbols	λ_{partial}	Memorials change, curricula do not; one channel of six is rewritten.
“Reverse racism” framing	η	The manufactured fear is doubled, not lowered; dissolution stays armed.
Date-limited remorse	r	The institutions re-encode the lie once the campaign ends; the memory resets to its written baseline.

Proposition 4 (Why every minimal intervention fails). *Let an armed lie be met by a counter-optimization that moves the subset $S \subsetneq \{\lambda, r, \alpha, \theta, \eta, \Theta_{\text{other}}\}$ of parameters, leaving the remainder at their racist values. Then the configuration (9) ultimately reasserts itself: the unchecked parameters resume their program because the writing that sustains them was not stopped.*

Proof. The fate of the system is a function of the full vector, not of any single component. Undisturbed, the remaining components evolve under their own laws (1)–(4): with the institutional writing r untouched the lie returns to L^* on the timescale set by the re-encoding; with θ untouched the channel stays closed and the account can still not be closed; with α untouched the current J remains one-way. The moved components relax back onto the moved-by- writing values. Only a full-vector reverse optimization has a stable fixpoint; every partial one is a temporary excursion. $\square$

Corollary 4 (The test of a real cure). *A real cure therefore has a distinguishing, measurable signature: the institutions change their writing, not the citizens their self-image. If, six months after the campaign, a school textbook, a statute book, a newspaper, and a street-name registry have all changed — the program is running. If only the language of the campaigners has changed, the program is running in reverse.*

Corollary 5 (No one is improved by this theory). *The corollary is uncomfortable in exactly the democratic direction. The framework does not reward the good conscience of the well-meaning; it rewards the auditing and rewiring of the machinery of memory. A citizen with a perfect heart inside an unchanged textbook is still a citizen of a racist configuration — and the theory says the configuration, not the heart, is the thing that kills.*

12 The Modern Record

The theory does not live only in the archives. The same audit of channels produces the modern matrix, and the same theorem predicts what happens to a society that runs the (9) configuration with modern instrumentation.

Table 9: The modern racism matrix: contemporary systems read as channel audits.

Contemporary system	The written lie	Active channel
Mass incarceration	<i>the target is criminal by nature</i>	law, media, language
Wealth gap	<i>generations of written exclusion need not be repaid</i>	law, finance, media
Health disparities	<i>the poorer biology of the target explains it</i>	science, media
Criminal-justice bias	<i>guilt by likeness</i>	law, media
Housing redlining	<i>the target lowers the value of the street</i>	finance, law, media

Remark 6. *Notice what is written and what is not. In each modern case the lie has retreated from the open language of the target is not human into the safer language of the gap is a fact of nature — measured, therefore real, therefore no one’s doing. The Θ_{other} erosion is the same program, run with the word measurement doing the work that inferiority once did. That is why the channel of science is listed in Table 1: a measuring instrument that points at the target’s outcomes and never at the machinery that produced them is a re-encoding instrument.*

13 The Complete IAR Theorem of Racism

We can now assemble the whole theory as a single statement.

Theorem 4 (The Complete IAR Theorem of Racism). *Racism is a physical configuration of an open system, not a moral accident. It is completely described by the vector (9):*

$$Racism = \Theta_{other \rightarrow 0} \oplus \lambda \text{ weaponized} \oplus \alpha \rightarrow 1 \oplus \theta \rightarrow 0, \quad (10)$$

subject to the following laws, each proven above:

1. **[The young lie dies.]** *An unwritten lie decays under λ and vanishes within a generation (Proposition 1).*
2. **[The written lie persists.]** *A lie rewritten by the channels of memory is fixed at $L^* = 1 - \lambda/r > 0$ and cannot decay (Proposition 2).*
3. **[The lie builds the program.]** *The six parameters are moved by the same dynamics: $\alpha \rightarrow 1$, $\theta \rightarrow 0$, $\eta \rightarrow 0.43$, $\Theta_{other} \rightarrow 0$ (Theorem 1).*

4. *[The armed lie is a war-seed.]* When $\Theta_{\text{other}} < 0.25$, $\theta < 0.30$, and $\eta > 0.30$ hold together, the system is armed; within four years of continued encoding it blooms (Scenario 3).
5. *[War is the collapse of theory of mind.]* The killing is the application of the lie under $\Theta_{\text{other}} = 0$ (Theorem 2).
6. *[The mirror is the cure.]* The reverse optimization makes the same constants work against the lie: an armed seed is neutralized within about a decade of institutional truth (Theorem 3, Scenario 5).

Thus: without encoding into memory, racism dies; with encoding, it is fixed and blooms into war; and the cure is to decode the memory, humanize the other, rebalance power, and reopen the channel — because the truth is the only quantity in the whole theory that a racist program cannot weaponize.

Remark 7. *The last clause deserves its own remark. A great deal of the theory’s difficulty comes from the fact that the lie can infiltrate every quantity of the system — including α , θ , and η . But the truth cannot be recruited to the lie’s side of a channel audit in the long run: truth is the only configuration that survives contact with the world. What the theory calls the truth is not a faction. It is the property of a memory whose writing survives verification. Decoding is therefore not a battle between two lies; it is a return of the memory to the standard of verification, and the standard is the same for the racist and for the healed society: keep only what survives contact with the world.*

14 Conclusion

This paper has given racism a physical theory. A lie was defined as a false belief engineered to occupy memory; the Young Lie was shown to decay, the Written Lie to persist; the persistence of the written lie was shown to move the six IAR parameters of a society into the configuration (9); that configuration was shown to be a war-seed, armed when the perceived humanity of the target collapses and the channel of exchange closes; and the historical and modern records were shown to support the prediction. The engine reproduced the whole chain in three scenarios from one law, and the counter-optimization was derived as the mechanical reverse.

What the theory does not do is comfortable. It says the difference between a prejudice and a holocaust is the local writing infrastructure; it says a racism is a memory operation, not a heart condition, so heart-changing alone has never and will never heal one; and it says the cost of healing is paid by whoever holds the power over memory. But the same theory gives the only hopeful claim in the field: the machinery is mechanical, and mechanical things can be reversed. The lie takes twenty-five years to build. The truth takes eight to disarm it. The account can be closed. The system can be rebalanced. Θ_{other} can come back to one.

Final motto: nothing in history was unavoidable that could instead be un-written. The racist program is a configuration of six numbers. Configurations can be changed.

A The Companion Engine

The companion program `code/iar_racism.py` implements Model 0 (1)–(4), the lie’s dynamical law (7), the derived-parameter laws of Section 5, and the arming and blooming conditions of the fate table. It uses only `numpy` and is fully deterministic; Section 6 reproduces its output verbatim. The engine is a demonstration of the physics of direction, and no part of it is a device with effects on the world beyond its own print out.

- **Society** — a society as the three state variables plus the six parameters;
- **RacistLie** — the lie with its propagation strength **spread** and severity, growing logistically under writing (7) and decaying under λ and any truth injection τ ;
- **run_decay** — Scenario A, $r = 0$;
- **run_encode** — Scenario B, a planted seed rewritten every cycle;
- **run_healing** — Scenario C, the reverse optimization;
- **iar_step** — the discrete form of (1)–(4).

To reproduce the three scenarios: `python3 code/iar_racism.py`.